

THIRANEX

DATA CLEANING & REPORTING AUTOMATION

Project Type: ETL Data Engineering & Automated Reporting | Internship Assignment 04

📊 DATA ENGINE PREPROCESSING METRICS (PIPELINE SUMMARY)

- **Automation Mechanism:** Python Pandas Script Orchestration
- **Execution Status:** 100% Automated Success

Pipeline Audit Metric	Raw State Count	Cleaned Post-Process Count	Resolved Action Executed
Duplicate Records	2 Rows Identified	1 Row Deduplicated	Redundant duplicate row dropped instantly via unique key matching.
Missing Values	3 Null Fields Detected	0 Null Fields Remaining	Imputed data gaps using numerical rolling group median tracking.
Inconsistent Formats	Mixed Case Text & Extra Spaces	Uniform Title-Cased Text	Cleansed whitespace padding and standardized text data cases.

📁 AUTOMATED EXECUTIVE REPORTING SUMMARY

Data Engineering Preprocessing Insights

- **Reporting Efficiency Boost:** Manual processing downtime reduced from hours to zero. The script extracts, transforms, and loads (ETL) data values instantly.
- **Database Accuracy Rating:** 100% integrity score achieved across all processed structural elements.

Business Category Revenue Highlights

- **Electronics Portfolio:** Dominates the commercial landscape with a total performance aggregate of **\$2,700**.
- **Office Supplies & Furniture:** Stable downstream sales pipelines totaling **\$1,350** and **\$900** respectively, providing sound core operational cash baselines.

